# Predictive Value of Biological Sex for Determining Level of Care in Symptomatic Poisoned Patients

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## Key Results for Abstract

### 1. Demographics (modeling subset)

**Original external validation set (n=105):** 56 females (53%), median age 28 [IQR 18–44]; 49 males (47%), median age 31 [IQR 22–50]; 21 ICU admissions (20%)

**Larger data set (n=189):** 101 females (53%), median age 29 [IQR 20–47]; 88 males (47%), median age 32 [IQR 26–51]; 38 ICU admissions (20%)

### 2. Sex-Exposure Association

**Overall chi-squared test: $p = 0.007$**

- Antidepressants: 78% female, 22% male
- Sedatives: 79% female, 21% male
- Analgesic: 57% female, 43% male
- Street Drugs: 35% female, 65% male
- CO, As, CN: 39% female, 61% male

### 3. Model Performance

**Original validation set → tested on accrued cases (n=84):**

| | Sensitivity | Specificity |
|---|---|---|
| Without sex | 0.118 | 0.985 |
| With sex | 0.353 | 0.970 |

**Larger data set (n=189, in-sample):**

| | Sensitivity | Specificity |
|---|---|---|
| Without sex | 0.289 | 0.987 |
| With sex | 0.342 | 0.967 |

## Sex-Exposure Table

Table 3: **Sex vs type of xenobiotic (N = 219, chi-squared p = 0.007).** Cells denote case count (row percentage).

| | Biological Sex | | | | | |
|---|---|---|---|---|---|---|
| | ***F*** | ***M*** | **Total** | **p-value** | **Row vs Rest p** | **Row vs Rest p (adj)** |
| Exposure Category | | | | | | |
| *Combination* | 23 (40%) | 34 (60%) | 57 (100%) | | 0.097 | 0.258 |
| *Analgesic* | 17 (57%) | 13 (43%) | 30 (100%) | | 0.611 | 0.698 |
| *CO, As, CN* | 11 (39%) | 17 (61%) | 28 (100%) | | 0.276 | 0.441 |
| *Unknown* | 15 (54%) | 13 (46%) | 28 (100%) | | 0.901 | 0.901 |
| *Antidepressants* | 21 (78%) | 6 (22%) | 27 (100%) | | **0.005** | **0.041** |
| *Street Drugs* | 7 (35%) | 13 (65%) | 20 (100%) | | 0.216 | 0.432 |
| *Alcohol* | 6 (40%) | 9 (60%) | 15 (100%) | | 0.555 | 0.698 |
| *Sedatives* | 11 (79%) | 3 (21%) | 14 (100%) | | 0.060 | 0.240 |
| Total | 111 (51%) | 108 (49%) | 219 (100%) | | | |

## Full Model Output

```
## === TRAINING: Model WITH sex (original dataset) ===

##
## Call:
## glm(formula = `Dichotomous Outcome` ~ `INTOXICATE SCORE` + Gender,
##     family = binomial, data = train_data)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -3.69013    0.72489  -5.091 3.57e-07 ***
## `INTOXICATE SCORE`  0.10777    0.04059   2.655  0.00793 **
## GenderM             1.89518    0.62540   3.030  0.00244 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 105.085  on 104  degrees of freedom
## Residual deviance:  84.422  on 102  degrees of freedom
## AIC: 90.422
##
## Number of Fisher Scoring iterations: 5

##
## Sex coefficient: 1.895 (95% CI: 0.752 to 3.260), p=0.002

## LRT p-value: < 0.001

##
## === TRAINING: Model WITH sex (combined dataset) ===

##
## Call:
## glm(formula = `Dichotomous Outcome` ~ `INTOXICATE SCORE` + Gender,
##     family = binomial, data = model_data)
##
## Coefficients:
##                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)        -3.50051    0.48586  -7.205 5.81e-13 ***
## `INTOXICATE SCORE`  0.13801    0.02988   4.618 3.87e-06 ***
## GenderM             1.29241    0.43178   2.993  0.00276 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 189.71  on 188  degrees of freedom
## Residual deviance: 151.27  on 186  degrees of freedom
## AIC: 157.27
##
## Number of Fisher Scoring iterations: 5

##
## Sex coefficient: 1.292 (95% CI: 0.471 to 2.178), p=0.003

##
## === AUC (for reference) ===
```

```
## Test set:  without sex 0.822, with sex 0.784

## Combined:  without sex 0.729, with sex 0.787
```